Lectures 1 and 2: Gaussians

Definition, linear structure, and conditioning as projection

OUTLINE $\varphi_X = \exp(\text{quadratic}) \longrightarrow AX+b$ is Gaussian $\longrightarrow$ uncorrelated $\Rightarrow$ independent $\longrightarrow$ conditioning is orthogonal projection $\longrightarrow$ ridge regression, and denoising

HOW TO READ Sections marked $*$ are not lectured. They are complete, they are examinable, and they carry most of the exercises. Exercises are typed A (did you follow the lecture), B (did you follow the argument) and C (one idea, short once you have it). Each week's quiz has one of each.

1 Gaussians, and their linear structure

1.1 Some facts from Fourier analysis

The following is standard, and it is the only material in the course we do not prove. Fix the convention now, since every sign below depends on it: the transform carries $e^{-ix\cdot\xi}$ and the inverse carries $e^{+ix\cdot\xi}$.

Proposition 1.1 (Fourier analysis, imported). *For $f \in L^1(\mathbb{R}^d)$ and a finite measure ν on $\mathbb{R}^d$ define*

$$\widehat{f}(\xi) \;=\; \int_{\mathbb{R}^d} f(x)\, e^{-ix\cdot\xi}\, dx, \qquad \widehat{\nu}(\xi) \;=\; \int_{\mathbb{R}^d} e^{-ix\cdot\xi}\, d\nu(x), \qquad \xi \in \mathbb{R}^d.$$

Then:

(i) ***Inversion.*** *If f and $\widehat{f}$ both lie in $L^1(\mathbb{R}^d)$ then $f(x) = (2\pi)^{-d}\int_{\mathbb{R}^d}\widehat{f}(\xi)\, e^{ix\cdot\xi}\, d\xi$ at every point of continuity of f.*

(ii) ***Derivatives.*** *If $f, \partial_j f \in L^1$ then $\widehat{\partial_j f}(\xi) = i\xi_j \widehat{f}(\xi)$; if $x_j f \in L^1$ then $\partial_{\xi_j}\widehat{f}(\xi) = (\widehat{-ix_j f})(\xi)$. Differentiating in x is multiplication by $i\xi_j$, and multiplication by x_j is $i\partial_{\xi_j}$.*

(iii) ***Isometry.*** *For $f \in L^1 \cap L^2$ one has $\|\widehat{f}\|_{L^2}^2 = (2\pi)^d \|f\|_{L^2}^2$, and the transform extends to a bijection of $L^2(\mathbb{R}^d)$ onto itself.*

(iv) ***Uniqueness.*** *If μ and ν are probability measures on $\mathbb{R}^d$ with $\widehat{\mu} = \widehat{\nu}$, then $\mu = \nu$.*

Part (iv) is what lets a law be *defined* by its transform, which is what we do next, and it is used a second time before this lecture ends. Part (ii) is the engine of almost every Gaussian computation in the course; part (i) is used once, in Lectures 3–4.

1.2 The definition

Definition 1.2 (Characteristic function). The *characteristic function* of a random vector X in $\mathbb{R}^d$ with law ν is the Fourier transform of ν:

$$\varphi_X(\xi) \;:=\; \widehat{\nu}(\xi) \;=\; \mathbb{E}\big[e^{-iX\cdot\xi}\big], \qquad \xi \in \mathbb{R}^d.$$

It is defined for every X, with no integrability assumption, because $|e^{i\theta}| = 1$. Immediately $\varphi_X(0) = 1$, $|\varphi_X(\xi)| \le 1$ and $\varphi_X(-\xi) = \overline{\varphi_X(\xi)}$.

Definition 1.3 (Gaussian vector)**.** Let $\mu \in \mathbb{R}^d$ and let Σ be a symmetric $d \times d$ matrix. A random vector X in $\mathbb{R}^d$ is *Gaussian with mean μ and covariance Σ*, written $X \sim \mathcal{N}(\mu, \Sigma)$, if

$$\boxed{\varphi_X(\xi) \;=\; \exp\Big(-i\,\mu\cdot\xi \;-\; \tfrac{1}{2}\,\xi^\top\Sigma\,\xi\Big)} \qquad \text{for every } \xi \in \mathbb{R}^d.$$

By Proposition 1.1(iv) at most one law satisfies this. Whether any does is the business of §1.3, and the answer will be: exactly when $\Sigma \in \mathrm{Sym}_{\geq 0}(\mathbb{R}^d)$, so nothing is lost by reading the definition with that restriction imposed from the start. First, the two words in the definition have to be earned.

Proposition 1.4 (The parameters are what they are called)**.** *If $X \sim \mathcal{N}(\mu, \Sigma)$ then $\mathbb{E}|X|^2 < \infty$, and*

$$\mathbb{E}[X_j] = \mu_j, \qquad \mathbb{E}\big[(X_j - \mu_j)(X_k - \mu_k)\big] = \Sigma_{jk}.$$

In particular the law determines (μ, Σ), so two Gaussians agree in law if and only if their parameters agree.

Proof. Write $\psi(\xi) = -i\mu\cdot\xi - \frac{1}{2}\xi^\top\Sigma\xi$, so $\varphi_X = e^\psi$ is smooth and

$$\partial_j\varphi_X\big|_{\xi=0} = -i\mu_j, \qquad \partial_j\partial_k\varphi_X\big|_{\xi=0} = \big(\partial_j\partial_k\psi + \partial_j\psi\,\partial_k\psi\big)\big|_{\xi=0} = -\Sigma_{jk} - \mu_j\mu_k.$$

Finiteness of the second moment is not an assumption; it is forced by this smoothness. For $h > 0$,

$$\frac{2\varphi_X(0) - \varphi_X(he_j) - \varphi_X(-he_j)}{h^2} \;=\; \mathbb{E}\Big[\frac{2 - 2\cos(hX_j)}{h^2}\Big],$$

and the left side converges to $-\partial_j^2\varphi_X(0) = \Sigma_{jj} + \mu_j^2$ as $h \downarrow 0$. The integrand is non-negative and converges pointwise to X_j^2, so Fatou gives $\mathbb{E}[X_j^2] \leq \Sigma_{jj} + \mu_j^2 < \infty$.

With $\mathbb{E}|X|^2 < \infty$ the difference quotients of $e^{-iX\cdot\xi}$ are dominated by $|X_j|$ and by $|X_jX_k|$, so we may differentiate under the expectation twice: $\partial_j\varphi_X(0) = -i\,\mathbb{E}[X_j]$ and $\partial_j\partial_k\varphi_X(0) = -\mathbb{E}[X_jX_k]$. Comparing with the display gives $\mathbb{E}[X_j] = \mu_j$ and $\mathbb{E}[X_jX_k] = \Sigma_{jk} + \mu_j\mu_k$. ■

Theorem 1.5 (Affine images)**.** *If $X \sim \mathcal{N}(\mu, \Sigma)$ and $A \in \mathbb{R}^{m\times d}$, $b \in \mathbb{R}^m$, then*

$$AX + b \;\sim\; \mathcal{N}\big(A\mu + b,\; A\Sigma A^\top\big).$$

Proof. $\varphi_{AX+b}(\xi) = e^{-ib\cdot\xi}\,\mathbb{E}\big[e^{-iX\cdot(A^\top\xi)}\big] = e^{-ib\cdot\xi}\varphi_X(A^\top\xi)$. Substituting Definition 1.3 and using $\mu\cdot A^\top\xi = (A\mu)\cdot\xi$ gives $\exp\big(-i(A\mu+b)\cdot\xi - \frac{1}{2}\xi^\top A\Sigma A^\top\xi\big)$. ■

Corollary 1.6 (Rotational invariance)**.** *If $X \sim \mathcal{N}(0, I_d)$ and $O \in O(d)$ then $OX \overset{d}{=} X$. More generally, for $X \sim \mathcal{N}(0, \Sigma)$ one has $OX \overset{d}{=} X$ precisely when $O\Sigma O^\top = \Sigma$.*

Proof. Theorem 1.5 gives $OX \sim \mathcal{N}(0, O\Sigma O^\top)$, and by Proposition 1.4 that law equals $\mathcal{N}(0, \Sigma)$ exactly when $O\Sigma O^\top = \Sigma$. For $\Sigma = I_d$ this is $OO^\top = I_d$. ■

Invariance is what singles the Gaussian out among product measures — Exercise 1.9 makes that exact. In November, invariance under orthogonal *conjugation* will single out a matrix ensemble in the same way.

Corollary 1.7 (One coordinate at a time). *$X \sim \mathcal{N}(\mu, \Sigma)$ if and only if $v \cdot X \sim \mathcal{N}(v \cdot \mu,\ v^\top \Sigma v)$ for every $v \in \mathbb{R}^d$.*

Proof. Forward: take $A = v^\top$ in Theorem 1.5. Conversely, the characteristic function of the scalar $v \cdot X$ evaluated at 1 is $\mathbb{E}[e^{-i\, v \cdot X}] = \varphi_X(v)$; by hypothesis it equals $\exp(-i\, v \cdot \mu - \frac{1}{2} v^\top \Sigma v)$, which is Definition 1.3. ■

Exercise 1.8 (TYPE A). Let $X \sim \mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^2$ with $\mu = (1, -1)$ and $\Sigma = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

(a) Write $\varphi_X(\xi)$ out explicitly and recover $\mathbb{E}[X_1 X_2]$ from it by differentiating, as in Proposition 1.4.

(b) Use Theorem 1.5 to find the laws of $X_1 + X_2$, of $X_1 - X_2$, and of the pair $(X_1 + X_2,\ X_1 - X_2)$. Are the two independent?

(c) Now let Σ be an arbitrary covariance on $\mathbb{R}^2$. For which Σ are $X_1 + X_2$ and $X_1 - X_2$ independent, and for which is there a $v \neq 0$ with $v \cdot X$ almost surely constant?

Exercise 1.9 (TYPE C). Let $d \geq 2$ and let X have independent and identically distributed coordinates with $OX \overset{d}{=} X$ for every $O \in O(d)$. Show that $X \sim \mathcal{N}(0, cI_d)$ for some $c \geq 0$. You may use that a characteristic function is continuous.

This is the sentence above, proved: rotational invariance and a product structure together admit nothing but the Gaussian. No moment assumption is needed, and none should be made.

1.3 Existence

Nothing so far exhibits a single Gaussian. One computation supplies all of them.

Lemma 1.10. *Let g have density $(2\pi)^{-1/2} e^{-x^2/2}$ on $\mathbb{R}$. Then $\mathbb{E}\big[e^{-i\lambda g}\big] = e^{-\lambda^2/2}$; that is, $g \sim \mathcal{N}(0, 1)$.*

Proof. Set $f(\lambda) = \mathbb{E}[e^{-i\lambda g}]$. Differentiating under the expectation,

$$f'(\lambda) \;=\; \frac{-i}{\sqrt{2\pi}} \int_{\mathbb{R}} x\, e^{-i\lambda x}\, e^{-x^2/2}\, dx.$$

The observation that does the work is $x\, e^{-x^2/2} = -\frac{d}{dx} e^{-x^2/2}$, which converts the factor of x into a derivative and lets integration by parts move it onto the other factor:

$$f'(\lambda) \;=\; \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\lambda x}\, \frac{d}{dx}\Big(e^{-x^2/2}\Big)\, dx \;=\; \frac{-i}{\sqrt{2\pi}} \int_{\mathbb{R}} \Big(\frac{d}{dx} e^{-i\lambda x}\Big) e^{-x^2/2}\, dx \;=\; (-i)(-i\lambda)\, f(\lambda) \;=\; -\lambda f(\lambda).$$

With $f(0) = 1$ this forces $f(\lambda) = e^{-\lambda^2/2}$. ■

Remark 1.11. What the proof used is $\mathbb{E}[g\, h(g)] = \mathbb{E}[h'(g)]$, with $h(x) = e^{-i\lambda x}$: against a standard Gaussian, multiplication by g acts like differentiation. Remember the move. It returns in Lecture 4 under its own name, and again in Lecture 5, where it proves Wick's theorem.

If $Z = (g_1, \ldots, g_d)$ has independent standard normal coordinates then

$$\varphi_Z(\xi) = \prod_{j=1}^{d} \mathbb{E}\big[e^{-i\xi_j g_j}\big] = \prod_{j=1}^{d} e^{-\xi_j^2/2} = e^{-|\xi|^2/2},$$

so $Z \sim \mathcal{N}(0, I_d)$: the standard Gaussian in d dimensions is a tensor product of one-dimensional ones. Every other Gaussian is an affine image of it.

Proposition 1.12 (Well-posedness). *Let $\mu \in \mathbb{R}^d$ and let Σ be symmetric. A random vector satisfying Definition 1.3 exists if and only if $\Sigma \succeq 0$, and is then unique in law. If $\Sigma = AA^\top$ one may take $X = AZ + \mu$ with $Z \sim \mathcal{N}(0, I_d)$.*

Proof. If $\Sigma \succeq 0$, factor $\Sigma = AA^\top$. Theorem 1.5 applied to $Z \sim \mathcal{N}(0, I_d)$ gives $AZ + \mu \sim \mathcal{N}(\mu, AA^\top) = \mathcal{N}(\mu, \Sigma)$. Uniqueness in law is Proposition 1.1(iv).

Conversely, suppose $v^\top \Sigma v = -c$ with $c > 0$. Along $\xi = sv$ the right side of Definition 1.3 has modulus $e^{s^2 c/2} \to \infty$, contradicting $|\varphi_X| \le 1$. ■

The family of Gaussian laws on $\mathbb{R}^d$ is therefore indexed exactly by $\mathbb{R}^d \times \mathrm{Sym}_{\ge 0}(\mathbb{R}^d)$, and the representation $X = AZ + \mu$ is a construction inside a proof rather than a rival definition.

Example 1.13 (The degenerate case). Take $d = 2$, $\mu = 0$ and $\Sigma = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, positive semidefinite of rank one. By Proposition 1.12 the law $\mathcal{N}(0, \Sigma)$ exists, and $A = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ realizes it as $X = (g, g)$. It has no density — it is supported on a line — and Definition 1.3 never notices. A definition through the density $\propto e^{-\frac{1}{2}x^\top \Sigma^{-1} x}$ would have had to exclude it, and there is nothing pathological about it to exclude.

Exercise 1.14 (TYPE A). Let $f(x) = (2\pi)^{-d/2} e^{-|x|^2/2}$ on $\mathbb{R}^d$. Show $\widehat{f}(\xi) = e^{-|\xi|^2/2}$, then verify Proposition 1.1(i) on this pair by evaluating $(2\pi)^{-d} \int_{\mathbb{R}^d} e^{-|\xi|^2/2} e^{ix\cdot\xi} d\xi$ and recovering f. Conclude that the standard Gaussian density is a fixed point of the Fourier transform up to the factor $(2\pi)^{d/2}$.

Exercise 1.15 (TYPE B). Let $\Sigma \succ 0$, write $\Sigma = AA^\top$ with A invertible, and let $X = AZ + \mu$.

(a) Change variables in the density of Z to obtain $p_X(x) = (2\pi)^{-d/2} (\det \Sigma)^{-1/2} \exp\big(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1} (x - \mu)\big)$, and check that the answer does not depend on which square root A was chosen.

(b) Show that when Σ is singular no density exists: produce $v \ne 0$ with $v \cdot (X - \mu) = 0$ almost surely, and conclude that X lives on an affine subspace of dimension less than d.

Part (a) is the definition most students meet first. Part (b) is what it cannot say, and Example 1.13 is the smallest instance.

1.4 Uncorrelated versus independent

For general random variables zero covariance says very little. For jointly Gaussian ones it says everything.

Theorem 1.16. *Let (X, Y) be jointly Gaussian in $\mathbb{R}^d \times \mathbb{R}^{d'}$ — the concatenated vector is Gaussian — with $\mathrm{Cov}(X, Y) = 0$. Then $X \perp\!\!\!\perp Y$.*

Proof. Write $\xi = (s, u)$. The covariance of the concatenated vector is block diagonal, so $\xi^\top \Sigma \xi = s^\top \Sigma_{XX} s + u^\top \Sigma_{YY} u$ and $\mu \cdot \xi = \mu_X \cdot s + \mu_Y \cdot u$. Hence

$$\varphi_{(X,Y)}(s, u) \;=\; \varphi_X(s)\, \varphi_Y(u),$$

which is the characteristic function of the product law $\mathcal{L}(X) \otimes \mathcal{L}(Y)$. By Proposition 1.1(iv) the joint law *is* that product. ■

Uniqueness has now done two different jobs in one lecture. The hypothesis that fails in practice is joint Gaussianity, not Gaussianity of the marginals.

Example 1.17 (Uncorrelated, standard normal, not independent)**.** Let $g \sim \mathcal{N}(0,1)$ and let ε be an independent random sign; set $X = g$ and $Y = \varepsilon g$. Conditionally on $\varepsilon = \pm 1$ we have $Y = \pm g \overset{d}{=} g$, so both marginals are standard normal. They are uncorrelated, since $\mathbb{E}[XY] = \mathbb{E}[\varepsilon]\mathbb{E}[g^2] = 0$. And $|X| = |Y|$ always, so X determines Y up to a sign.

Conditioning on ε and applying Lemma 1.10,

$$\varphi_{(X,Y)}(s,u) \;=\; \tfrac{1}{2}e^{-(s+u)^2/2} + \tfrac{1}{2}e^{-(s-u)^2/2},$$

a sum of two Gaussian transforms and not exp(quadratic). The pair fails Definition 1.3 even though each coordinate satisfies it.

Exercise 1.18 (TYPE B)**.** Continue Example 1.17.

(a) Find the law of $X+Y$ and show it is not Gaussian, by exhibiting an atom.

(b) Corollary 1.7 says a vector is Gaussian as soon as every linear functional of it is. Compute the law of $sX + uY$ for arbitrary s, u and say why there is no contradiction.

(c) Does (X,Y) have a density on $\mathbb{R}^2$?

1.5 $*$ Entropy, and why the Gaussian maximizes it

Not lectured. Exercises due at the start of Lecture 2.

Definition 1.3 is not the only characterization of the Gaussian. Among all laws on $\mathbb{R}^d$ with a given mean and covariance $\Sigma \succ 0$, it is the one of largest differential entropy — the least committal law consistent with a second moment. That is what this section proves, and Lecture 3 is where it gets used: it is the answer to *why a Gaussian at all.*

Before entropy, surprise. If an event of probability p occurs, the surprise $s(p)$ ought to satisfy $s(1) = 0$ and $s(pq) = s(p)+s(q)$. Substituting $p = e^{-u}$ turns the second into $g(u+v) = g(u)+g(v)$ for $g(u) = s(e^{-u})$, whose only continuous solutions are linear. So $s(p) = -c\log p$, the base of the logarithm is a choice of units, and we take $c = 1$.

The **entropy** of a law on a finite set is the surprise expected before looking,

$$H(p) = \mathbb{E}\big[-\log p(X)\big] = -\sum_x p(x)\log p(x),$$

and for a density p on $\mathbb{R}^d$ the same formula defines the **differential entropy** $h(p) = -\int p\log p$. The second is not the first in disguise, and the next exercise is where that becomes visible.

Exercise 1.19 (TYPE A)**.** (a) Compute H for a coin of bias q; sketch it on $q \in [0,1]$ and find where it is largest. Compute H for the uniform law on n points.

(b) Compute h for the uniform law on $[0,a]$, the exponential of rate λ, and $\mathcal{N}(0,\sigma^2)$ — you should find $\frac{1}{2}\log(2\pi e\sigma^2)$ for the last. Exhibit a law with *negative* differential entropy, and say why no discrete entropy can be negative.

(c) Show $h(aX) = h(X) + \log|a|$ for $a \neq 0$. So "the entropy of a length" is not a well-defined number, while H has no units at all.

The pattern we are after is this: *fix a constraint, and entropy is maximized by the least committal law satisfying it.* With no constraint on a finite set that is the uniform law. The Gaussian will answer the same question with a different constraint — but part (c) shows h is the wrong quantity to state the answer with, and the cure is to measure entropy *relative* to a reference. For densities p, q, the **relative entropy** is $D(p \,\|\, q) = \int p \log(p/q)$: the extra surprise from expecting q when the truth is p, so it should never be negative.

Exercise 1.20 (TYPE B). Prove **Gibbs' inequality**: $D(p \,\|\, q) \geq 0$, with equality iff $p = q$. *Jensen applied to* $-\log$. Deduce the discrete instance $H(p) \leq \log n$ for every law on n points, with equality only for the uniform one, by taking q uniform. Then check that $D(p \,\|\, q)$ is unchanged when X is rescaled — unlike h.

Now fix μ and $\Sigma \succ 0$, let q be the Gaussian density, and let p be any density with that mean and covariance. Expanding $0 \leq D(p \,\|\, q)$ gives $0 \leq -h(p) - \int p \log q$, and everything hinges on the second term.

Exercise 1.21 (TYPE B). Write out $\log q$ and observe it is a *quadratic polynomial* in x. Deduce that $\int p \log q$ depends on p only through its mean and covariance, hence equals $\int q \log q = -h(q)$. Conclude $h(p) \leq h(q)$ with equality only when $p = q$, and check $h(q) = \frac{1}{2} \log \big((2\pi e)^d \det \Sigma \big)$.

Note how little the Gaussian had to do: everything turned on $\log q$ being quadratic, so that integrating it against p could see only p's first two moments. *A quadratic is exactly what a covariance constraint measures.*

Exercise 1.22 (TYPE C). Let X, Y be i.i.d., *symmetric*, each with a density. Show

$$h\Big(\frac{X+Y}{\sqrt{2}}\Big) \;\geq\; h(X).$$

You may use subadditivity $h(U,V) \leq h(U) + h(V)$, which is Gibbs' inequality applied to a joint density against the product of its marginals. Lecture 3 opens with this inequality.

Exercise 1.23 (TYPE C). Fix $S \subseteq \mathbb{R}^d$, functions $T_1, \ldots, T_k$ on S and numbers $c_1, \ldots, c_k$. **Assume** that for some $\lambda_1, \ldots, \lambda_k$ the function $\exp(\sum_m \lambda_m T_m)$ has finite integral Z over S, so that $q = Z^{-1} \exp(\sum_m \lambda_m T_m)$ is a density, and that $\int q T_m = c_m$ for each m. Show q uniquely maximizes h among densities on S with those moments.

Then reread the paragraph above — *which property of the Gaussian did that argument use?* Recover as instances the uniform law on $[0, a]$, the exponential at fixed mean on $[0, \infty)$, and $\mathcal{N}(\mu, \Sigma)$ at fixed mean and covariance. Finally, show the assumption does real work: on $S = \mathbb{R}$ with the mean alone constrained no λ makes $e^{\lambda x}$ integrable, and there is no maximizer — exhibit densities of mean 0 with h arbitrarily large.

2 Conditioning as projection

> **Q.** We observe Y. What is our best guess for X?
>
> Before answering we must say what *best* means.

Fix jointly Gaussian (X, Y) in $\mathbb{R}^p \times \mathbb{R}^m$, both centered for readability. Nothing below needs $p = 1$: projection acts on one coordinate at a time, and we write ΠX for the vector of projections of the coordinates of X.

2.1 What would count as an answer

Measure a guess by mean squared error. A predictor is any function of the observation, and we want the one minimizing $\mathbb{E}|X - f(Y)|^2$.

Definition 2.1 (Conditional expectation)**.** $\mathbb{E}[X \mid Y]$ is the minimizer of $\mathbb{E}|X - f(Y)|^2$ over all measurable $f : \mathbb{R}^m \to \mathbb{R}^p$ with $\mathbb{E}|f(Y)|^2 < \infty$.

One could instead define $\mathbb{E}[X \mid Y]$ as the orthogonal projection of X onto the span of the coordinates of Y. That would make Theorem 2.3 a tautology. Under Definition 2.1 the competitors are all functions of Y, including nonlinear ones.

2.2 The right inner product

On the linear span of the coordinates of X and Y — a finite-dimensional space of *scalar* random variables — put

$$\langle U, V \rangle \;:=\; \mathbb{E}[UV].$$

Inner product is covariance; orthogonal is uncorrelated. Let $H_Y = \operatorname{span}\{Y_1, \dots, Y_m\}$ and let Π be orthogonal projection onto H_Y.

Lemma 2.2 (Best linear predictor)**.** *$\Pi X = \Sigma_{XY}\Sigma_{YY}^{-1}Y$, with $\Sigma_{XY} = \mathbb{E}[XY^\top] \in \mathbb{R}^{p\times m}$ and $\Sigma_{YY} = \mathbb{E}[YY^\top]$. If Σ_{YY} is singular the same formula holds with a pseudo-inverse, and ΠX remains unique.*

Proof. Take $p = 1$; the general case is this one applied to each coordinate. Write $\Pi X = \sum_j c_j Y_j$. Orthogonality of the residual to each Y_k reads $\mathbb{E}[(X - \sum_j c_j Y_j)Y_k] = 0$, that is $\Sigma_{YY}c = \Sigma_{YX}$. ■

Lemma 2.2 holds for any square-integrable pair.

2.3 The theorem

Theorem 2.3 (Conditioning is projection)**.** *Let (X, Y) be jointly Gaussian and centered. Then*

$$\mathbb{E}[X \mid Y] \;=\; \Pi X \;=\; \Sigma_{XY}\Sigma_{YY}^{-1}Y.$$

Proof. Let $R = X - \Pi X$. By construction $\operatorname{Cov}(R, Y) = \Sigma_{XY} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YY} = 0$, and (R, Y) is a linear image of (X, Y), hence jointly Gaussian by Theorem 1.5. So Theorem 1.16 gives $R \perp\!\!\!\perp Y$. For any competitor f,

$$\mathbb{E}\big|X - f(Y)\big|^2 = \mathbb{E}|R|^2 + \mathbb{E}\big|\Pi X - f(Y)\big|^2 + 2\,\mathbb{E}\big[\langle R, \Pi X - f(Y)\rangle\big].$$

The cross term vanishes: $\Pi X - f(Y)$ is a function of Y, and R is independent of Y with mean zero, so the expectation factors. Hence $\mathbb{E}|X - f(Y)|^2 \geq \mathbb{E}|R|^2$, with equality exactly when $f(Y) = \Pi X$ almost surely. ■

Remark 2.4. Uncorrelatedness of R and Y does not suffice to kill the cross term; independence does, and independence comes from Theorem 1.16.

2.4 The conditional law

Since $X = \Pi X + R$ with $R \perp\!\!\!\perp Y$, conditioning on $Y = y$ leaves R untouched.

Corollary 2.5 (Schur complement)**.** *For jointly Gaussian* (X, Y)*,*

$$X \mid Y = y \;\sim\; \mathcal{N}\Big(\mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y), \;\; \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}\Big).$$

Remark 2.6*.* The conditional covariance does not depend on y.

Remark 2.7*.* Conditioning on $\{Y = y\}$ conditions on an event of probability zero. The decomposition $X = \Pi X + R$ with $R \perp\!\!\!\perp Y$ constructs a conditional law directly, so Corollary 2.5 is a definition justified by Theorem 2.3.

2.5 What the geometry gives in general

Everything so far has been Gaussian. The projection picture does not need that, and three facts we will use later fall out of it directly. Throughout this subsection X is any square-integrable scalar random variable and no joint Gaussianity is assumed.

Lemma 2.8 (Orthogonality characterizes it)**.** $m(Y) = \mathbb{E}[X \mid Y]$ *if and only if*

$$\mathbb{E}\big[(X - m(Y))\, f(Y)\big] = 0 \qquad \textit{for every } f \textit{ with } \mathbb{E}f(Y)^2 < \infty.$$

Proof. For any competitor, $\mathbb{E}[(X - f(Y))^2] = \mathbb{E}[(X - m(Y))^2] + 2\mathbb{E}[(X - m(Y))(m(Y) - f(Y))] + \mathbb{E}[(m(Y) - f(Y))^2]$. If the middle term is always 0 then m is the minimizer. Conversely, if $\mathbb{E}[(X - m(Y))h(Y)] = c \neq 0$ for some h with $\mathbb{E}h(Y)^2 = 1$, then $f = m + ch$ gives $\mathbb{E}[(X - f(Y))^2] = \mathbb{E}[(X - m(Y))^2] - c^2$, so m was not minimal. ∎

That the residual is orthogonal to *every* function of Y — not merely to the coordinates of Y — is the whole difference between Definition 2.1 and projection onto a span. The next three statements are read off it.

Corollary 2.9. *Let* Z *be a further observation. Then, writing* $\mathbb{E}[X \mid Y, Z]$ *for the conditional expectation given the pair,*

$$\mathbb{E}\big[\,\mathbb{E}[X \mid Y, Z] \,\big|\, Y\,\big] \;=\; \mathbb{E}[X \mid Y] \qquad \textit{(the } \text{tower property}\textit{)},$$

and $\mathbb{E}\big[g(Y)X \mid Y\big] = g(Y)\,\mathbb{E}[X \mid Y]$ *for bounded* g*.*

Proof. A function of Y is in particular a function of (Y, Z). So for any such f, Lemma 2.8 applied twice gives $\mathbb{E}[Xf(Y)] = \mathbb{E}\big[\mathbb{E}[X \mid Y, Z]f(Y)\big] = \mathbb{E}\big[\mathbb{E}[\mathbb{E}[X \mid Y, Z] \mid Y]f(Y)\big]$, and the outer object satisfies the characterization of $\mathbb{E}[X \mid Y]$. For the second claim, $\mathbb{E}[(g(Y)X - g(Y)\mathbb{E}[X \mid Y])f(Y)] = \mathbb{E}[(X - \mathbb{E}[X \mid Y])\,(gf)(Y)] = 0$. ∎

Geometrically the tower property is one sentence: *functions of* Y *form a subspace of functions of* (Y, Z)*, and projecting onto a subspace of a subspace is projecting onto the smaller one.* It is what Lecture 4 uses to pass from conditioning on $X_1, \dots, X_n$ to conditioning on their sum, and it is the definition of a martingale in Module II.

Corollary 2.10 (Law of total variance)**.** *With* $\mathrm{Var}(X \mid Y) := \mathbb{E}\big[(X - \mathbb{E}[X \mid Y])^2 \,\big|\, Y\big]$*,*

$$\mathrm{Var}(X) \;=\; \mathbb{E}\big[\,\mathrm{Var}(X \mid Y)\big] \;+\; \mathrm{Var}\,\big(\mathbb{E}[X \mid Y]\big).$$

Proof. Centre and split: $X - \mathbb{E}X = (X - \mathbb{E}[X \mid Y]) + (\mathbb{E}[X \mid Y] - \mathbb{E}X)$. The second term is a function of Y, so by Lemma 2.8 the two are orthogonal, and the identity is Pythagoras. ■

Unexplained variation plus explained variation, and the decomposition is a right angle.

Exercise 2.11 (TYPE A)**.** Show $\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X]$ and $\mathrm{Var}\big(\mathbb{E}[X \mid Y]\big) \le \mathrm{Var}(X)$, and say exactly when the second is an equality and when the two sides of Corollary 2.10 put all the variance in the first term.

Exercise 2.12 (TYPE B)**.** Let J be a fair coin and, given J, let $X \sim \mathcal{N}(\pm 1, \sigma^2)$ accordingly. Compute both terms of Corollary 2.10 with $Y = J$ and check the total against a direct computation of $\mathrm{Var}(X)$. Then suppose only that V satisfies $\mathbb{E}[V \mid X_1, \dots, X_n] = -W/n$ for some W that is a function of $X_1, \dots, X_n$; deduce $\mathbb{E}[V \mid W] = -W/n$ and name the property used. *That deduction is exactly the step Lecture 4 needs to identify a Stein pair.*

2.6 Bayesian linear models and ridge regression

Take

$$y \;=\; Hw + \sigma\varepsilon, \qquad w \sim \mathcal{N}(0, \tau^2 I_p), \quad \varepsilon \sim \mathcal{N}(0, I_n) \text{ independent of } w.$$

The pair (w, y) is a linear image of (w, ε), hence jointly Gaussian by Theorem 1.5. Reading off blocks,

$$\Sigma_{ww} = \tau^2 I, \qquad \Sigma_{wy} = \tau^2 H^\top, \qquad \Sigma_{yy} = \tau^2 H H^\top + \sigma^2 I,$$

and Corollary 2.5 gives the posterior. Simplifying with $\tau^2 H^\top(\tau^2 H H^\top + \sigma^2 I)^{-1} = (H^\top H + \lambda I)^{-1} H^\top$:

Proposition 2.13. *With $\lambda = \sigma^2/\tau^2$,*

$$\mathbb{E}[w \mid y] = \big(H^\top H + \lambda I\big)^{-1} H^\top y, \qquad \mathrm{Cov}(w \mid y) = \sigma^2 \big(H^\top H + \lambda I\big)^{-1}.$$

The posterior mean is ridge regression, with regularization parameter the ratio of noise variance to prior variance. Three claims are now visible at once: *a loss function is a noise model, a regularizer is a prior, and fitting is conditioning.*

The exercises below are due at the start of Lecture 3.

Exercise 2.14 (TYPE A)**.** (a) Show $-\log p(y \mid w) = \frac{1}{2\sigma^2}\|y - Hw\|^2 + c$ with c free of w, so least squares is maximum likelihood. Say in one sentence what someone assumes about the world when they choose squared error, and whether they usually know it.

(b) Show the maximizer of the posterior density is $\widehat{w}_\lambda = \arg\min_w \|y - Hw\|^2 + \lambda\|w\|^2$. What does a large λ say about the data relative to the prior, and what are the limits $\tau \to 0$ and $\tau \to \infty$?

Exercise 2.15 (TYPE B)**.** Fill in the derivation above: compute the three blocks of the joint covariance and apply Corollary 2.5. Compare its length with completing the square in the density. Then note that $\mathrm{Cov}(w \mid y)$ does not depend on y, and say what that means operationally — the data came out surprising, so by how much should you revise your uncertainty?

The posterior mean is linear in y, so by Theorem 2.3 the best predictor of w happens to be linear. That is special.

Exercise 2.16 (TYPE B)**.** Let $Y \sim \mathcal{N}(0, 1)$ and $X = Y^2 - 1$. Show $\mathrm{Cov}(X, Y) = 0$, so the best *linear* predictor of X is the constant 0 with mean squared error 2. Exhibit a predictor with error 0. Which hypothesis of Theorem 2.3 fails?

2.7 * The two forms of the ridge estimator

Not lectured. Proposition 2.13 inverts a $p \times p$ matrix; the derivation that produced it inverted an $n \times n$ one. Both are correct.

Exercise 2.17 (TYPE C)**.** Show that for $\lambda > 0$

$$\big(H^\top H + \lambda I_p\big)^{-1} H^\top \;=\; H^\top \big(HH^\top + \lambda I_n\big)^{-1},$$

and deduce that ridge regression with $p \gg n$ costs $O(n^3)$ rather than $O(p^3)$. Then take $\tau \to \infty$ with $n < p$ and H of full row rank: show the posterior mean converges to $H^\top(HH^\top)^{-1}y$, the **minimum-norm** solution of $Hw = y$.

Ordinary least squares does not exist in this regime and the posterior does. The estimator you have just found is the one that interpolates the data, and Module II explains why its test error behaves as it does.

2.8 * Denoising is conditioning

Not lectured. Ridge regression looked backwards, at a classical estimator. The same theorem read forwards is the basis of every diffusion model, and the statement is worth having now even though Module II is where it is used.

Let X be any random vector in $\mathbb{R}^d$ with $\mathbb{E}|X|^2 < \infty$ — **no Gaussian assumption** — and observe it through Gaussian noise,

$$Y \;=\; X + \sigma\xi, \qquad \xi \sim \mathcal{N}(0, I_d) \text{ independent of } X.$$

Because the noise has a density, so does Y: $p_Y(y) = \mathbb{E}[\phi_\sigma(y - X)]$ with ϕ_σ the $\mathcal{N}(0, \sigma^2 I_d)$ density. The function $\nabla \log p_Y$ is called the *score* of Y.

Proposition 2.18 (Tweedie's formula)**.**

$$\mathbb{E}\big[X \mid Y\big] \;=\; Y + \sigma^2\, \nabla \log p_Y(Y).$$

Proof. By Lemma 2.8 it is enough to show $\mathbb{E}\big[\big(X - Y - \sigma^2 \nabla \log p_Y(Y)\big) f(Y)\big] \;=\; 0$ for every bounded smooth f, one coordinate at a time.

Two integrations by parts. First, conditionally on X the variable ξ is standard Gaussian, so the identity of Lemma 1.10 — $\mathbb{E}[\eta F(\eta)] = \mathbb{E}[F'(\eta)]$ for scalar standard Gaussian η, one integration by parts against $\eta\, e^{-\eta^2/2} = -\frac{d}{d\eta} e^{-\eta^2/2}$, applied in each coordinate — gives

$$\mathbb{E}\big[(Y - X) f(Y)\big] = \sigma\, \mathbb{E}\big[\xi f(X + \sigma\xi)\big] = \sigma^2\, \mathbb{E}\big[\nabla f(Y)\big].$$

Second, integrating by parts in y against the density of Y,

$$\mathbb{E}\big[\nabla f(Y)\big] = \int \nabla f(y)\, p_Y(y)\, dy = -\int f(y)\, \nabla p_Y(y)\, dy = -\,\mathbb{E}\big[f(Y)\, \nabla \log p_Y(Y)\big].$$

Combining, $\mathbb{E}[(Y - X) f(Y)] = -\sigma^2 \mathbb{E}[f(Y) \nabla \log p_Y(Y)]$, which rearranges to the claim. ■

The proof used one thing from Lecture 1 — that multiplication by a Gaussian acts like differentiation — and one thing from this lecture, that orthogonality to every test function *is* conditional expectation. Lecture 4 will give the first a name.

Exercise 2.19 (TYPE A)**.** Take $X \sim \mathcal{N}(0, \tau^2 I_d)$. Compute p_Y and $\nabla \log p_Y$ explicitly, and check that Proposition 2.18 returns $\mathbb{E}[X \mid Y] = \frac{\tau^2}{\tau^2+\sigma^2} Y$ — the same answer as Theorem 2.3. Which of the two derivations used that X was Gaussian?

Exercise 2.20 (TYPE B)**.** Justify the two exchanges of derivative and integral in the proof: show p_Y is smooth with $\nabla p_Y(y) = \mathbb{E}[\nabla \phi_\sigma(y - X)]$, and state what is needed for the boundary term in the second integration by parts to vanish. *Convolution with a Gaussian is what makes both steps free; a general Y need not have a differentiable density at all.*

Exercise 2.21 (TYPE C)**.** A *denoiser* is a function $D : \mathbb{R}^d \to \mathbb{R}^d$ trained by least squares, $D = \arg\min \mathbb{E}|X - D(Y)|^2$ over all D. Show that knowing D is the same as knowing $\nabla \log p_Y$: each determines the other explicitly. Conclude that fitting a denoiser to noisy samples — which requires no knowledge of p_X and no likelihood — **is** estimating the score of the noised law.

Then say in one sentence what this buys: given the score at every noise level σ, one can run the noising backwards. That is Module II.11, and it is a diffusion model.